

PROJECT WALKTHROUGH & LESSONS LEARNED

Inventory Mismatch Dispute Resolution — E-Commerce Supply Chain

Prepared by: Karthika Ayyanan | Version: v1.0 | Date: April 22, 2026 | Status: Draft

1. Project Overview

This project is about an inventory dispute between SnapMart, a mid-sized seller, and QuickKart, a large e-commerce platform. SnapMart dispatched 500 units to QuickKart's fulfillment warehouse, but QuickKart's system recorded only 320 units received — creating a discrepancy of 180 missing units. The project investigates where the 180 units went, identifies who is responsible, and recommends preventive measures to avoid recurrence.

Key Details

Domain	E-Commerce / Supply Chain Management
Case Study	Inventory Mismatch Dispute — SnapMart vs QuickKart
Role	Business Analyst
Actors	SnapMart, QuickKart, Logistics Partner, End Customer, BA
Tools Used	draw.io (diagrams), Microsoft Word (documents), Claude AI (learning support)
Duration	April 12 - April 22, 2026 (10 days)

2. Artifact Walkthrough

The following section documents each BA artifact built during this project — what was built, why it was needed, and the key insight gained from building it.

Artifact	What Was Built	Why It Was Needed	Key Learning
BRD	A 9-section Business Requirements Document capturing scope, stakeholders, 14 business requirements, 7 functional requirements, 6 non-functional requirements, assumptions, constraints, and risks	To establish a clear foundation for the investigation — defining what must be covered, who is involved, and what the system must do to resolve the dispute	A BRD is not just about listing requirements — the real skill is communicating effectively with stakeholders to understand their needs, gather the right requirements, and suggest improvements they may not have thought of

Business Use Case Diagram	A UML diagram showing 5 actors and 11 use cases within the Inventory Dispute Management System boundary	To provide a visual overview of who interacts with the system and what actions they perform — giving all stakeholders a shared understanding of the system scope	Building a Use Case Diagram is straightforward once you clearly understand your actors and their functions — the diagram almost draws itself when you know the system well
Process Flows (As-Is + To-Be)	Two swimlane process flow diagrams across 4 lanes (SnapMart, Logistics Partner, QuickKart, BA) — one showing the current broken process and one showing the improved future process	To visually identify exactly where the inventory mismatch occurs in the current process and to design a better process that prevents recurrence	The To-Be process was the most insightful part — it showed how a BA does not just document problems but actively designs solutions by reimagining the workflow from scratch
Stakeholder Matrix	A structured matrix mapping 8 stakeholders with their roles, power levels, interest levels, and engagement strategies using the Power/Interest framework	To ensure the right stakeholders are engaged in the right way throughout the investigation — preventing communication gaps and ensuring accountability	Every stakeholder has a different level of power and interest — understanding this helps a BA prioritize communication and engagement strategies effectively rather than treating all stakeholders the same
RACI Matrix	A responsibility matrix mapping 10 key tasks across 4 parties — defining who is Responsible, Accountable, Consulted, and Informed for each task	To eliminate confusion about roles and ownership — ensuring every task has clear accountability and no gaps in responsibility	Separating stakeholders into R, A, C, and I roles brings clarity to every task — ensuring no one is confused about their responsibility and no task falls through the cracks
Context Model	A high-level system diagram showing the Inventory Dispute Management System at the center with 5 external entities and all data flows between them	To provide a 30,000-foot view of the entire system — showing what data enters and exits the system from each external party	A Context Model gives a bird's eye view of the entire system — showing exactly what data flows between external entities and the system, making complex interactions simple to understand

3. Key Learnings

The following are the most important insights gained across the entire project — applicable to any BA engagement, not just this case study.

1. Requirements are conversations, not lists

The most important BA skill is not writing requirements — it is asking the right questions and listening carefully. Every requirement in the BRD started as a plain conversational answer that was then shaped into a professional statement.

2. Diagrams tell stories that documents cannot

Process flows, use case diagrams, and context models communicate complex information visually in seconds. A well-drawn As-Is/To-Be diagram shows the problem and solution simultaneously — something a written document cannot achieve as effectively.

3. Every stakeholder is different

The Stakeholder Matrix revealed that not all stakeholders should be engaged the same way. High-power stakeholders need to be managed closely while low-power stakeholders just need to be kept informed. Treating everyone the same wastes time and creates noise.

4. Accountability prevents chaos

The RACI Matrix showed that most project failures happen not because of lack of skills but because of unclear ownership. When every task has exactly one accountable party, there are no gaps, no blame games, and no confusion.

5. Start broad, then go deep

The correct BA order is Context Model first (big picture), then Use Case Diagram (who does what), then Process Flows (step by step detail). Understanding the sequence helps structure any BA engagement effectively.

6. As-Is before To-Be

It is impossible to design a better process without deeply understanding the current one. The As-Is process revealed exactly where the inventory mismatch occurred — which then directly informed every recommendation in the To-Be process.

4. Reflections

What I Would Do Differently

If starting this project again, I would spend more time upfront mapping out each stakeholder's role clearly before jumping into the RACI Matrix. Understanding who owns each task versus who just contributes to it was the trickiest part — and having a clearer stakeholder picture from the start would have made RACI assignment much smoother.

I would also follow the correct BA order from the beginning — starting with the Context Model before the BRD. Although building the BRD first helped me understand the case study deeply, presenting artifacts in the correct BA sequence (Context Model → Use Case → Process Flows → BRD) would have made the project flow more naturally.

Most Challenging Part

The most challenging part was understanding actor flows — specifically how information and responsibilities move between multiple parties across different artifacts. For example, drawing cross-lane arrows in process flows and correctly assigning R/A/C/I in the RACI Matrix both required careful thinking about who does what at each step. Working through this confusion, however, deepened my understanding of the case study significantly.

What This Project Taught Me About BA Work

- BA work is 80% thinking and communication, 20% documentation
- Every artifact has a purpose — none of them exist just to fill a portfolio
- The best BAs do not just document problems — they design solutions
- Stakeholder management is as important as requirements management
- Learning by doing is far more effective than learning by reading

ROOT CAUSE ANALYSIS (RCA)

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1. Problem Statement

SnapMart Sellers dispatched 500 units of inventory to QuickKart's fulfillment warehouse. However, QuickKart's system recorded only 320 units received — creating a discrepancy of 180 missing units. This has resulted in order failures, customer dissatisfaction, financial losses, and an active dispute between SnapMart and QuickKart over accountability for the missing inventory.

2. RCA Methodology

This Root Cause Analysis uses the 5 Whys technique — a systematic approach that asks 'Why?' repeatedly until the underlying root cause of a problem is identified, rather than stopping at surface-level symptoms.

Element	Details
Technique	5 Whys Analysis
Problem	180 unit inventory discrepancy between SnapMart and QuickKart
Scope	Specific shipment of 500 units dispatched by SnapMart in this dispute
Analyst	Karthika Ayyanan (Business Analyst)

3. Possible Cause Paths

Before applying the 5 Whys, three possible cause paths were identified for the 180 unit discrepancy:

Path	Cause	Description	Likelihood
A	SnapMart dispatched fewer units	SnapMart's system shows 500 but actual units packed may have been fewer due to data entry error or damaged stock	High
B	QuickKart warehouse miscount	QuickKart's barcode scanner missed units during inbound scanning or staff counted incorrectly	High
C	Lost during transit	Logistics partner lost, damaged or misdelivered 180 units during transportation	Medium

Note: Path C (Logistics) is the hardest to detect as there is no real-time tracking between handover points. However, Paths A and B are most probable as system and scanner errors are the most common causes of inventory discrepancies in e-commerce operations.

4. 5 Whys Analysis

The following 5 Whys chain traces the problem from its visible symptom to the underlying root cause:

PROBLEM: QuickKart's system shows only 320 units received against SnapMart's 500 units dispatched — a discrepancy of 180 units

Why 1	Why does QuickKart show only 320 units received?	Because either SnapMart dispatched fewer units than recorded OR QuickKart's warehouse system failed to log all received units accurately
Why 2	Why was the count not verified at the point of delivery?	Because no one from both parties was present to jointly count and confirm the units at the exact moment of handover — the logistics partner simply dropped off and left
Why 3	Why was there no joint verification at the handover point?	Because there was no mandatory joint verification process defined between SnapMart, the Logistics Partner, and QuickKart for inbound inventory handovers
Why 4	Why was there no mandatory joint verification process?	Because no standard GRN (Goods Received Note) process existed between both parties — there was no signed document confirming unit count at time of delivery
Why 5	Why was there no standard GRN process?	Because there was no formal inventory handover policy agreed upon between SnapMart and QuickKart — each shipment was handled inconsistently without a defined procedure

ROOT CAUSE

The absence of a standardized inventory verification and handover process between SnapMart, the Logistics Partner, and QuickKart — specifically the lack of a formal GRN process and joint count verification at the point of delivery.

5. Corrective Actions

The following corrective actions directly address the identified root cause and are already captured in the project BRD as Prevention Requirements:

BRD Ref	Why #	Corrective Action	Owner
PR-1	Why 3	QuickKart's warehouse team must physically count and verify all inbound inventory upon arrival; both parties must sign off on confirmed unit count	QuickKart
PR-3	Why 4	SnapMart must obtain a signed GRN from the logistics partner; QuickKart must verify inbound units against this GRN upon warehouse receipt	SnapMart + QuickKart
PR-4	Why 5	A standardized dispute resolution and inventory handover procedure must be documented and agreed upon by both parties	SnapMart + QuickKart
PR-2	Why 2	SnapMart must have real-time visibility of their inventory levels within QuickKart's fulfillment system at all times	QuickKart

6. Conclusion

The root cause of the 180 unit inventory discrepancy is not a simple human error or system glitch — it is a systemic process gap. The absence of a formal, standardized inventory verification policy meant that each shipment was handled inconsistently, making it impossible to detect or prove discrepancies at the point they occurred.

The corrective actions defined in the BRD — mandatory joint verification, GRN process, real-time inventory visibility, and a standardized dispute procedure — directly address each layer of the root cause chain and will prevent similar disputes from occurring in the future.